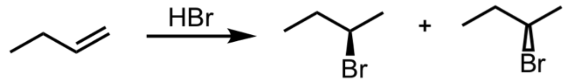
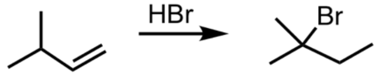
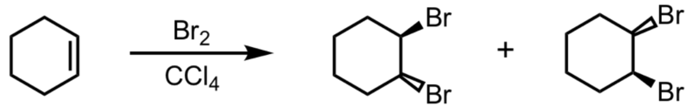
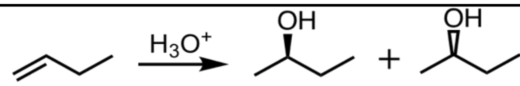
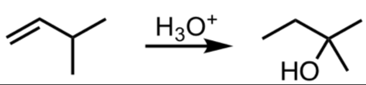
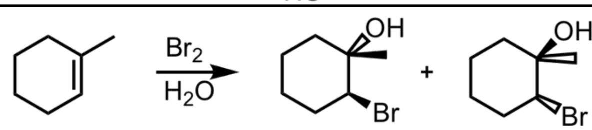
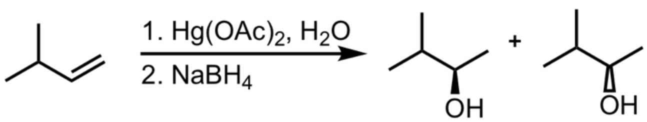
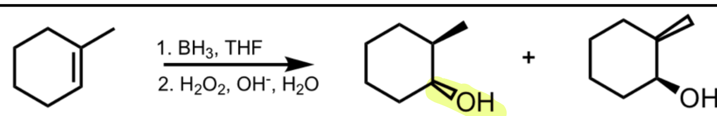
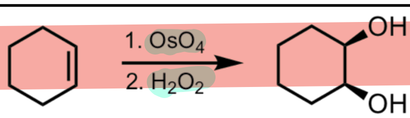
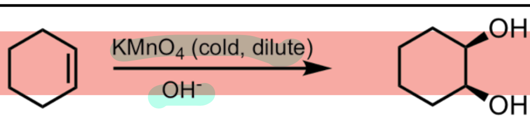
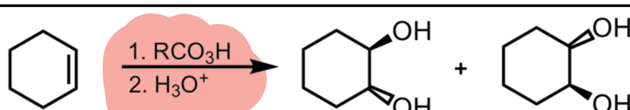
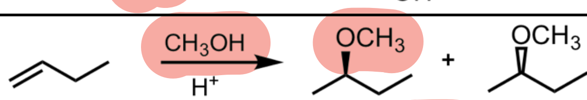
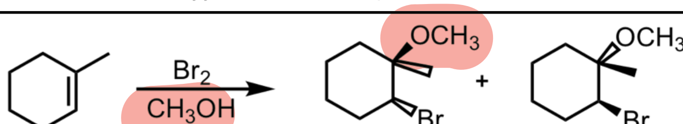
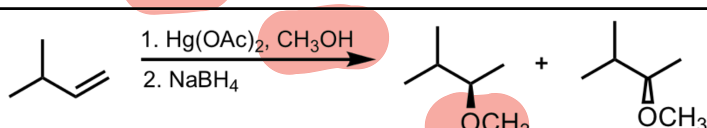
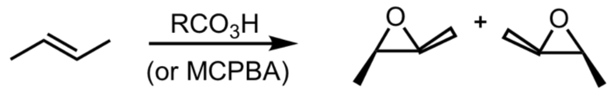
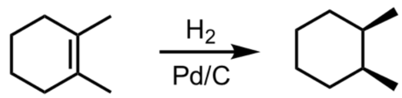
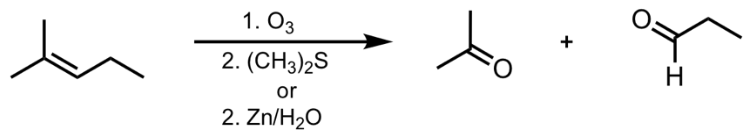
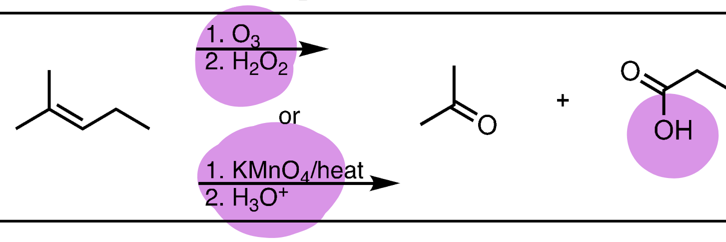
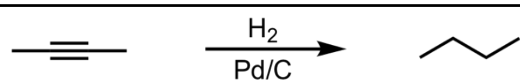
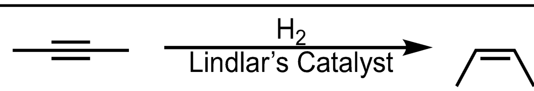
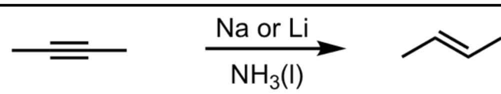
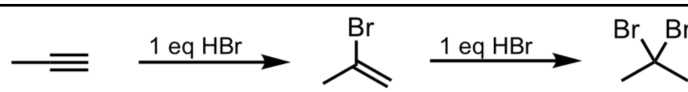
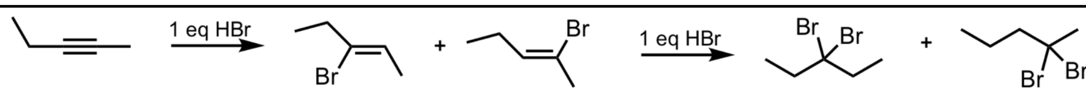
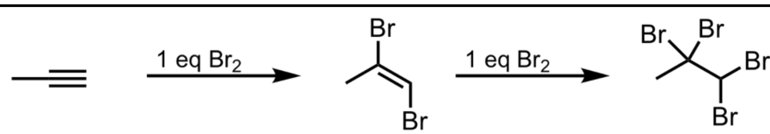
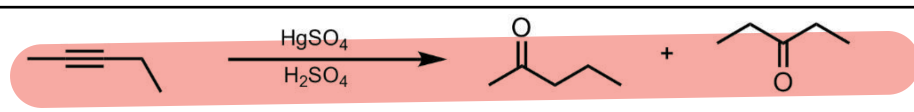
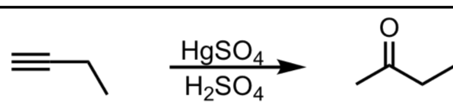
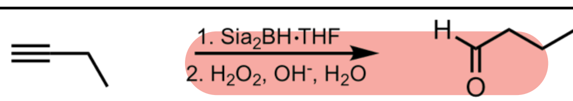
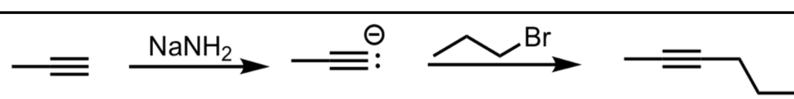
Bootcamp.com Organic Chemistry Reaction Summary Sheet

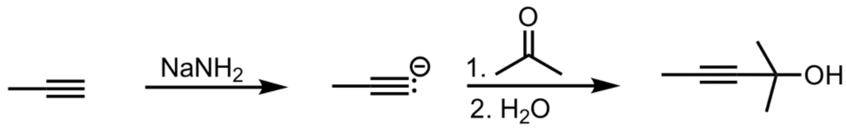
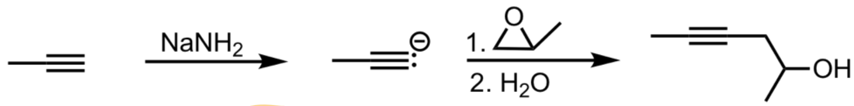
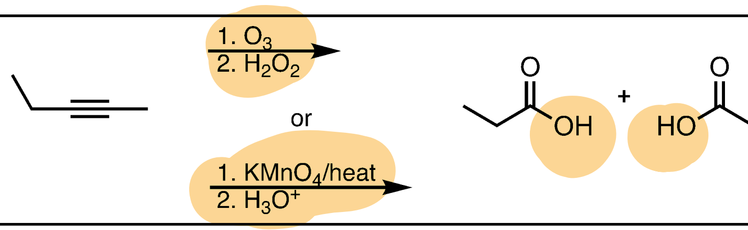
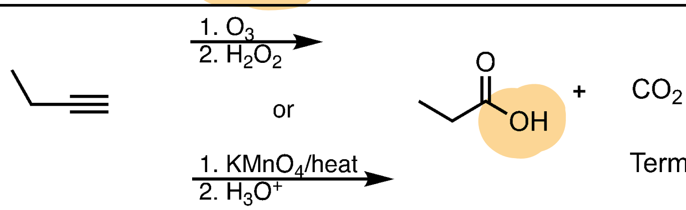
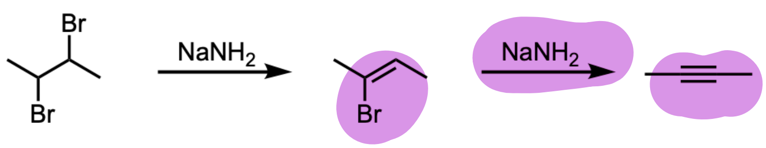
Alkene Reactions

Hydrohalogenation	
Hydrohalogenation (with Rearrangement)	
Halogenation	
Hydrobromination with Peroxide	
Hydration	
Hydration (with Rearrangement)	
Bromination in H_2O	
Oxymercuration-Demercuration	
Hydroboration-Oxidation	
Syn-Dihydroxylation	 meso
Syn-Dihydroxylation	 meso
Anti-Dihydroxylation	
Addition of an Alcohol	
Bromination in Alcohol	
Alkoxymercuration-Demercuration	

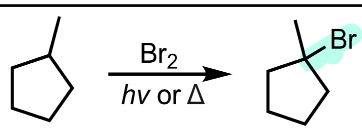
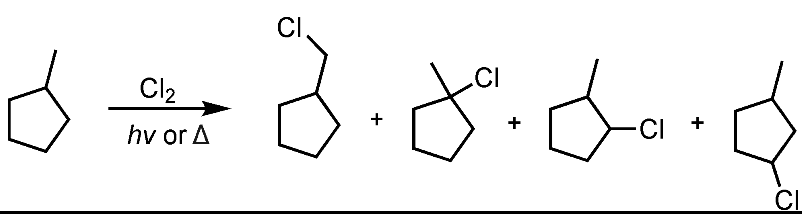
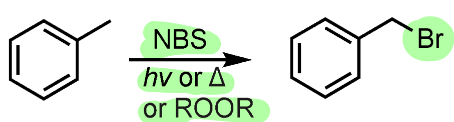
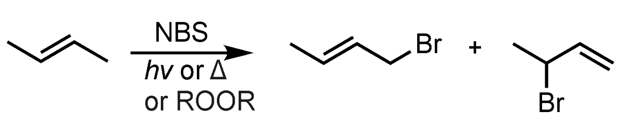
Epoxidation	
Catalytic Hydrogenation	 meso *Pt can also be used*
Ozonolysis (Reducing Conditions)	
Ozonolysis (Oxidizing Conditions)/Oxidative Cleavage	

Alkyne Reactions

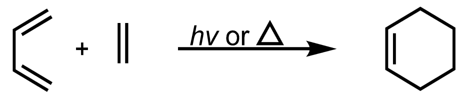
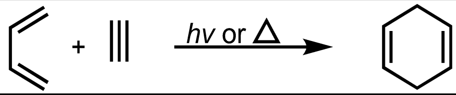
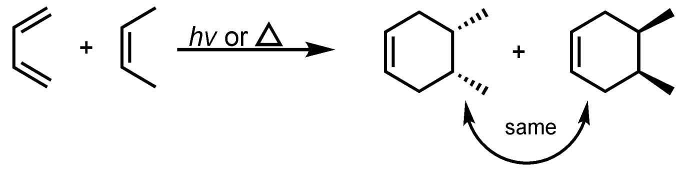
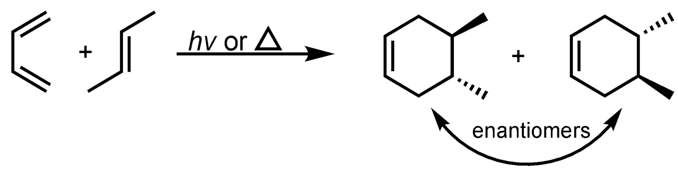
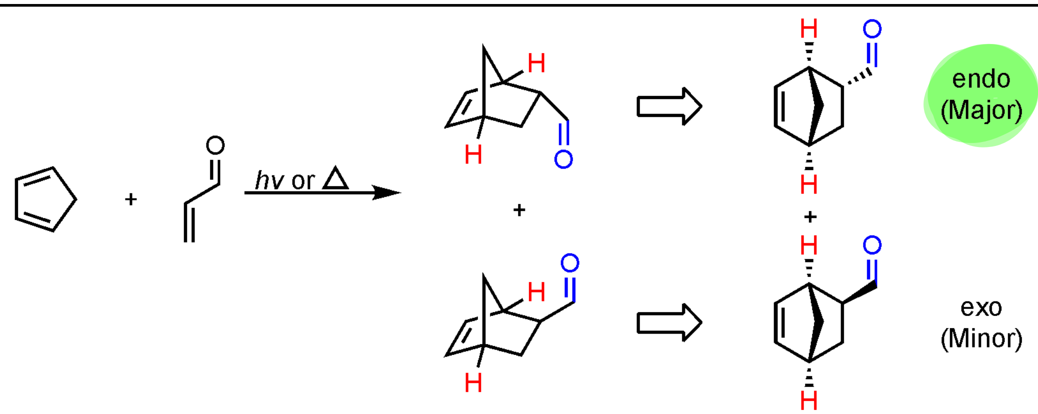
Catalytic Hydrogenation (Catalytic Reduction)	
Reduction to Cis-Alkene	
Reduction to Trans-Alkene	
Hydrohalogenation with HBr (Terminal Alkyne)	
Hydrohalogenation with HBr (Internal Alkyne)	
Halogenation with Br_2	
Hydration of an Internal Alkyne	
Hydration of a Terminal Alkyne (Markovnikov)	
Hydration of a Terminal Alkyne (Anti-Markovnikov)	
S_N2 Addition of an Acetylide Ion to an Alkyl Halide	

S_N2 Addition of an Acetylide Ion to a Ketone	
S_N2 Addition of an Acetylide Ion to an Epoxide	
Ozonolysis/Oxidative Cleavage on an Internal Alkyne	
Ozonolysis/Oxidative Cleavage on a Terminal Alkyne	 Terminal carbon becomes CO_2
Alkyne Formation from Double Elimination of a Vicinal Dihalide	

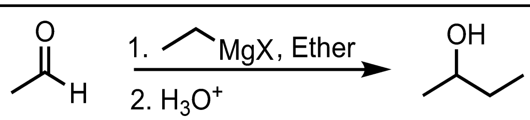
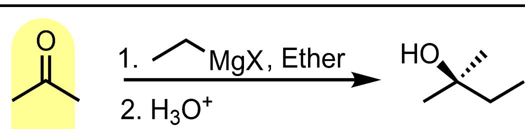
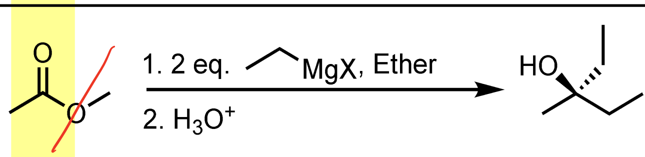
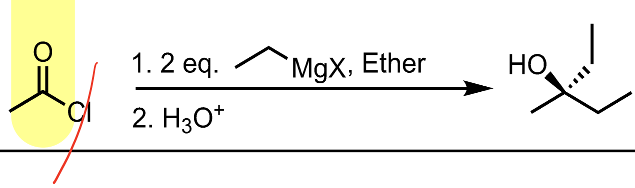
Free Radical Halogenation Reactions

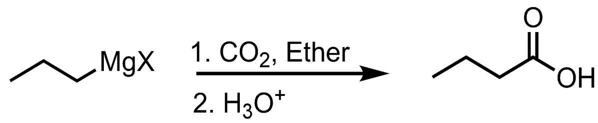
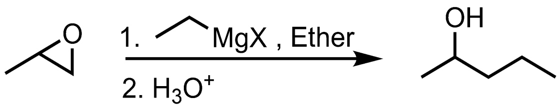
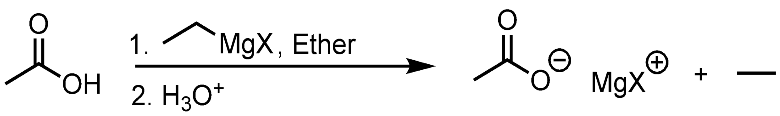
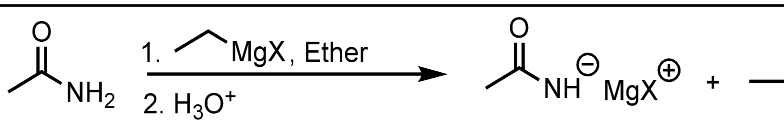
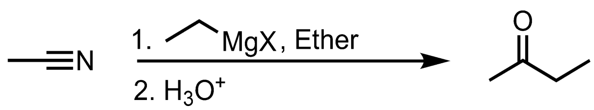
Free Radical Halogenation using Bromine (more selective)	
Free Radical Halogenation using Chlorine (less selective)	
Allylic/Benzylic Bromination	 Allylic 

Diels-Alder Reactions

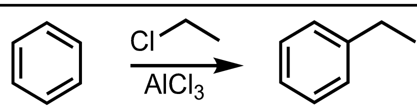
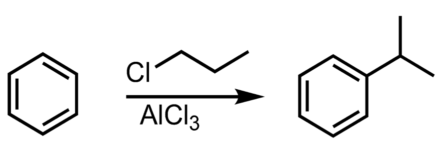
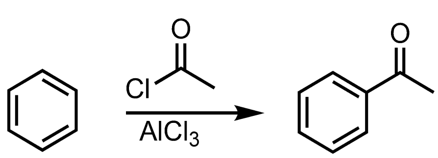
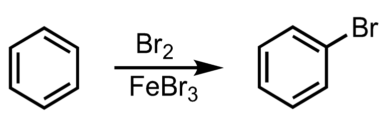
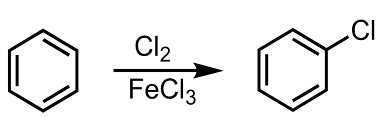
Diene Addition to a Dienophile (Alkene)	
Diene Addition to a Dienophile (Alkyne)	
Diene Addition to a *cis* Dienophile	
Diene Addition to a *trans* Dienophile	
Diene Addition to a substituted Dienophile	

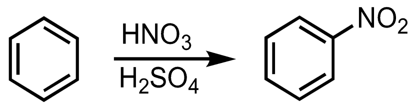
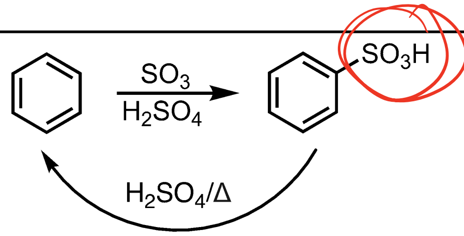
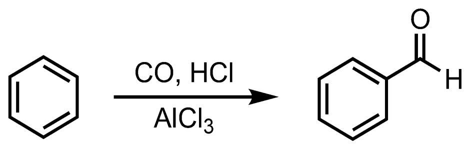
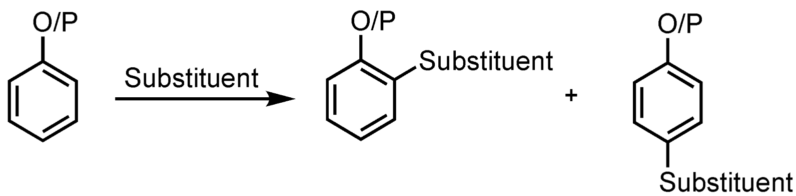
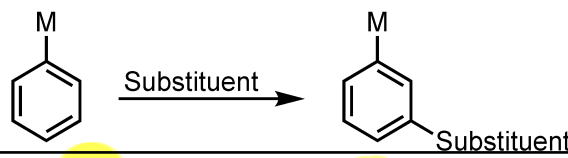
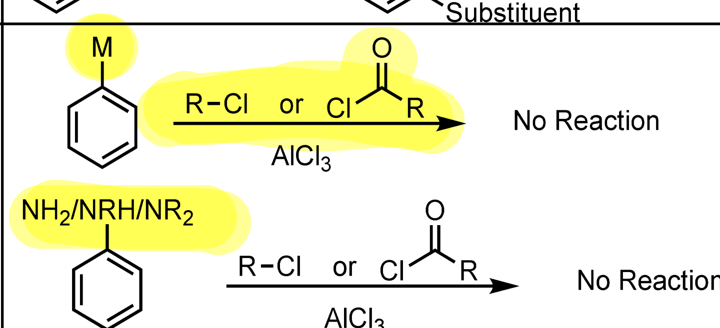
Grignard Reactions

Addition of a Grignard Reagent to an Aldehyde	 2˚Alcohol
Addition of a Grignard Reagent to a Ketone	 3˚Alcohol
Addition of a Grignard Reagent to an Ester	 3˚Alcohol
Addition of a Grignard Reagent to an Acyl Chloride	 3˚Alcohol

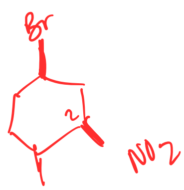
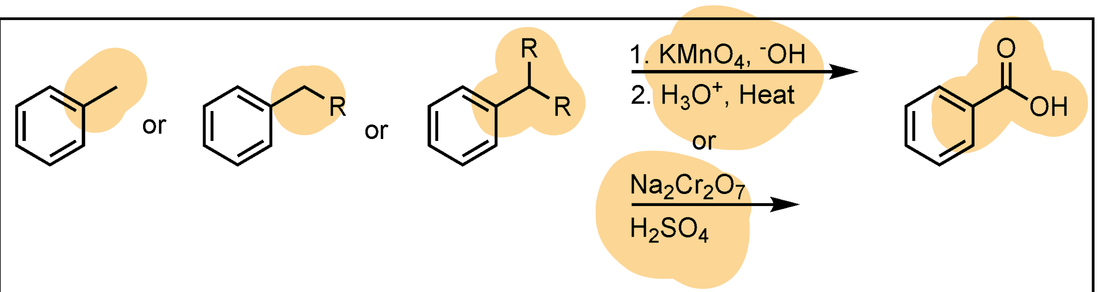
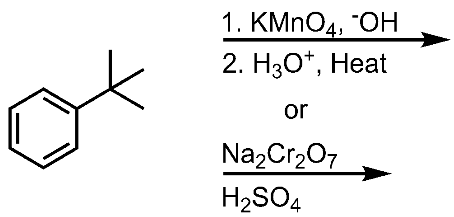
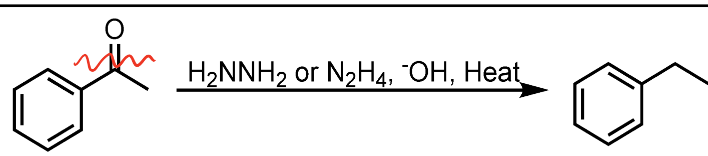
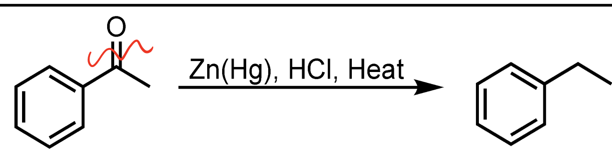
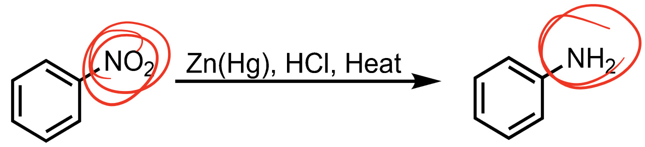
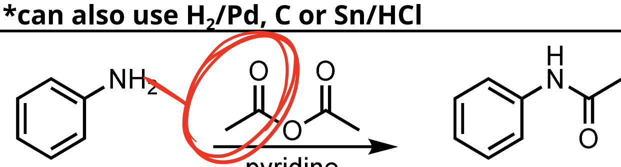
Addition of a Grignard Reagent to CO_2	 Carboxylic Acid
Addition of a Grignard Reagent to an Epoxide (adds to the less subs. side)	
Addition of a Grignard Reagent to a Carboxylic Acid	 Carboxylate
Addition of a Grignard Reagent to an Amide	 Deprotonated Amide
Addition of a Grignard Reagent to a Nitrile	 Ketone

Electrophilic Aromatic Substitution (EAS) Reactions

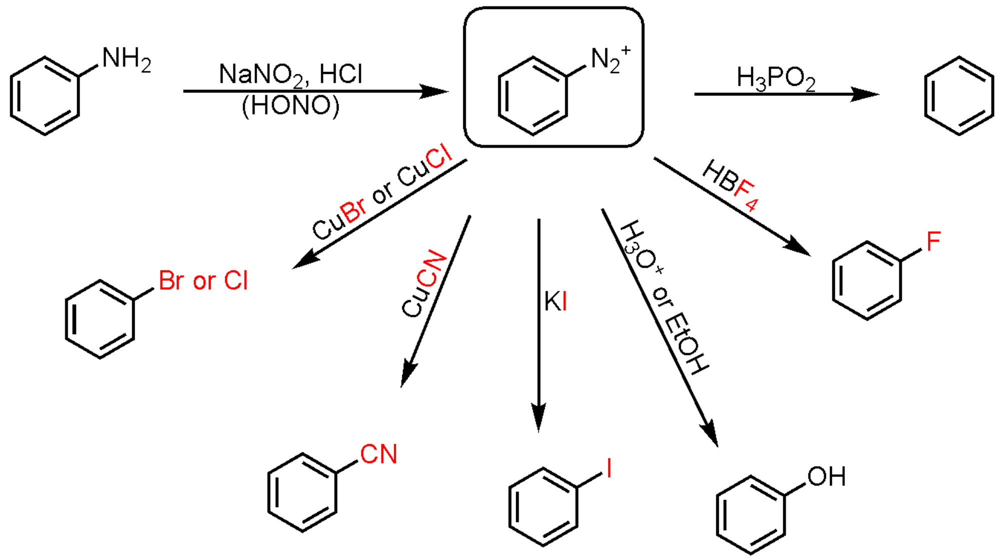
Friedel-Crafts Alkylation (Rearrangement Possible)	 
Friedel-Crafts Acylation (No Rearrangement Possible)	
Bromination	
Chlorination	

Nitration	
Sulfonation	
Formylation	
EAS with an ortho/para-directing group on Benzene	
EAS with a meta-directing group on Benzene	
Friedel-Crafts Alkylation/Acylation with a meta-directing group or an amine on Benzene	

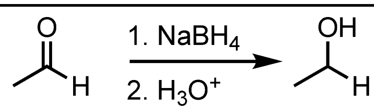
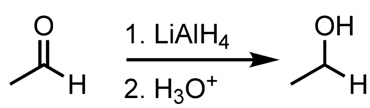
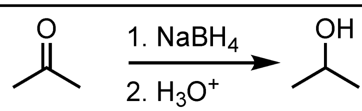
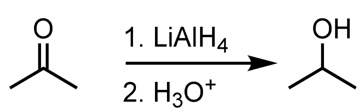
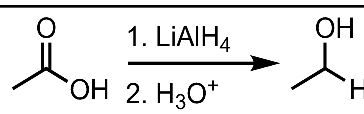
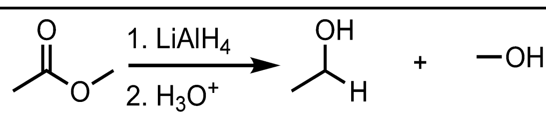
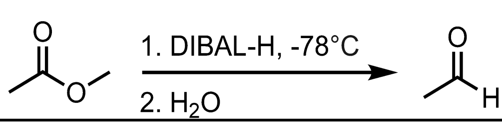
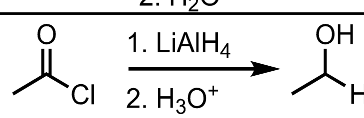
Benzene Side-Chain Reactions

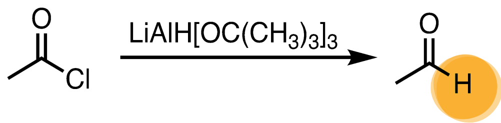
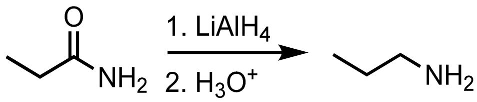
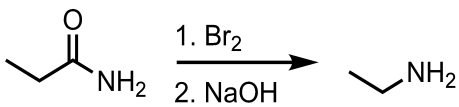
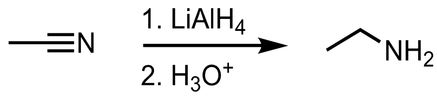
Side-Chain Oxidation of Benzene to form Benzoic Acid 	  No Reaction (Requires free Hydrogen at Benzylic position)
Wolff-Kishner Reduction	
Clemmensen Reduction	 ***can also use H_2/Pd, C**  ***can also use H_2/Pd, C or Sn/HCl**
Acetylation of Aniline using Acetic Anhydride	 Aniline Acetanilide

Diazonium Salt Reactions

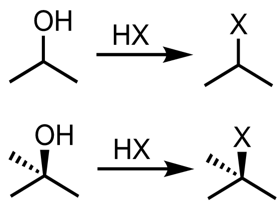
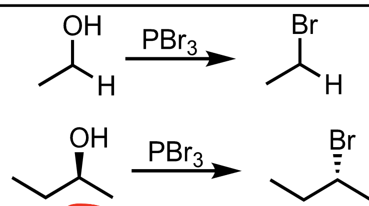
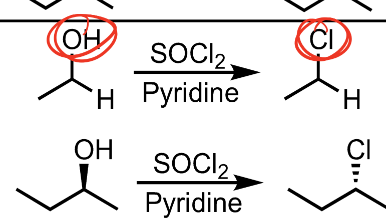
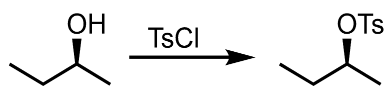
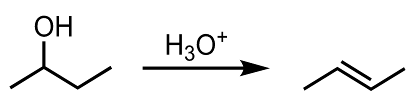
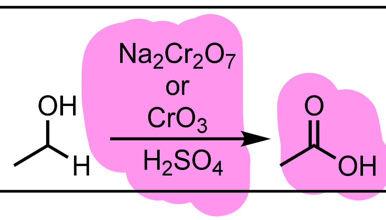


Hydride Reduction Reactions

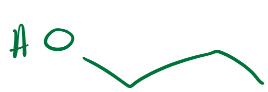
Reduction of an Aldehyde to a 1˚Alcohol	 
Reduction of a Ketone to a 2˚Alcohol	 
Reduction of a Carboxylic Acid to a 1˚Alcohol	
Reduction of an Ester to a 1˚Alcohol	
Reduction of an Ester to an Aldehyde	
Reduction of an Acyl Chloride to a 1˚Alcohol	

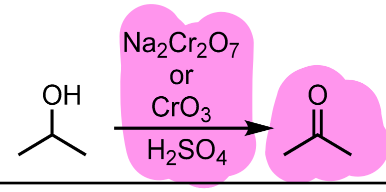
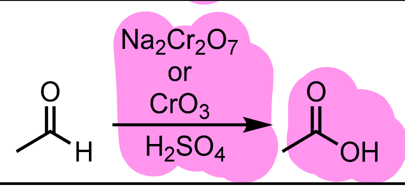
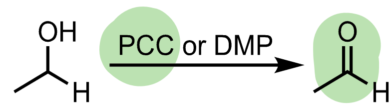
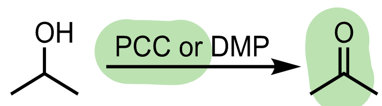
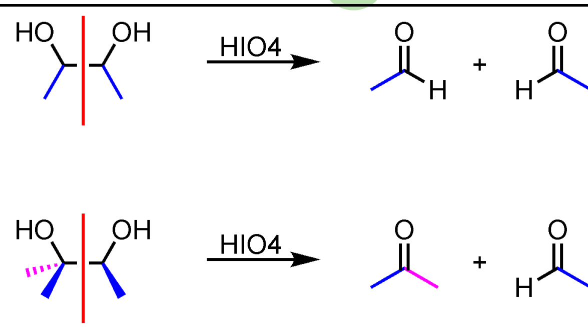
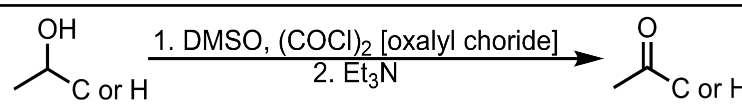
Reduction of an Acyl Chloride to an Aldehyde	
Reduction of an Amide to an Amine	
Hofmann Rearrangement	
Reduction of a Nitrile to an Amine	

Alcohol Reactions

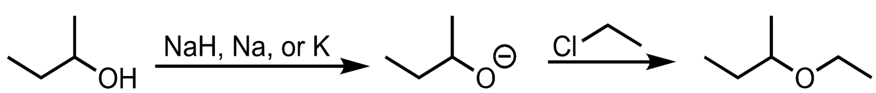
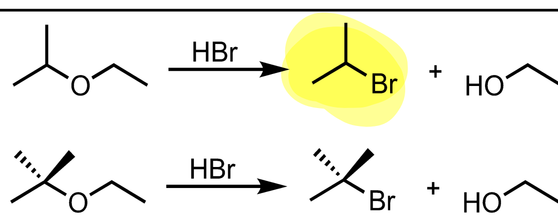
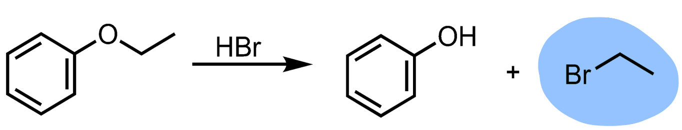
Conversion of a 2˚/3˚Alcohol to an alkyl halide via S_N1	
Conversion of a 1˚/2˚Alcohol to an alkyl bromide via S_N2	
Conversion of a 1˚/2˚Alcohol to an alkyl chloride via S_N2	
Conversion of an Alcohol to a Tosylate Ester (OTs)	 Retention of Stereochemistry
Acid-catalyzed Dehydration of an Alcohol	 Zaitsev's Rule
Chromic Acid Oxidation of a 1° Alcohol to a Carboxylic Acid	

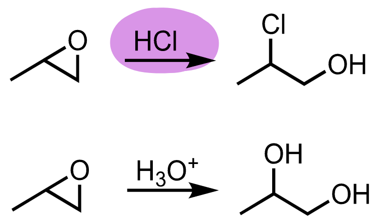
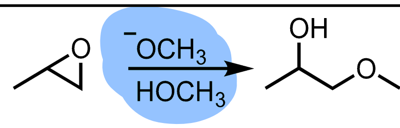
n- propanol



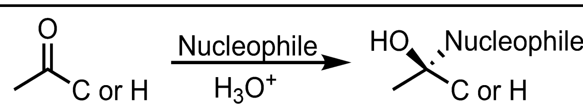
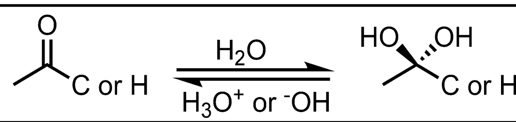
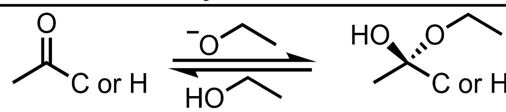
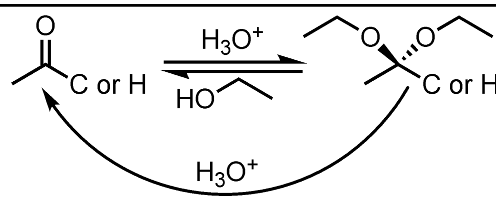
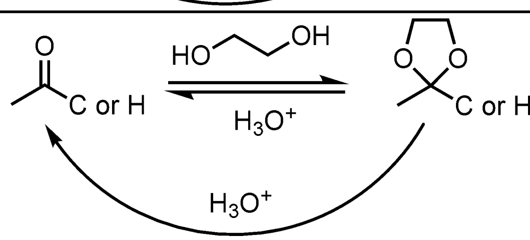
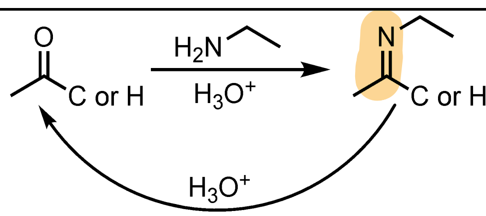
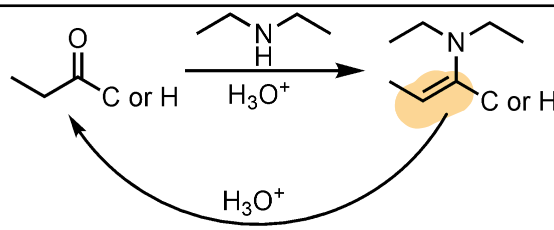
Chromic Acid Oxidation of a 2° Alcohol to a Ketone	
Chromic Acid Oxidation of an Aldehyde to a Carboxylic Acid	
PCC or DMP Oxidation of a 1° Alcohol to an Aldehyde	
PCC or DMP Oxidation of a 2° Alcohol to a Ketone	
Oxidative Cleavage of a 1,2 Diol	
Swern Oxidation	

Ether and Epoxide Reactions

Williamson Ether Synthesis via S_N2	
Acid-catalyzed Cleavage of Ethers when one side is 2˚/3˚ (Nucleophile attacks more substituted side via S_N1)	
Acid-catalyzed Cleavage of Ethers when neither side is 2˚/3˚ (Nucleophile attacks less substituted side via S_N2)	

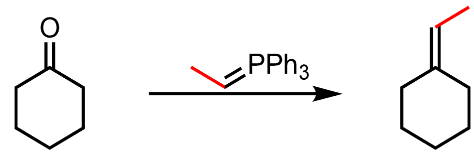
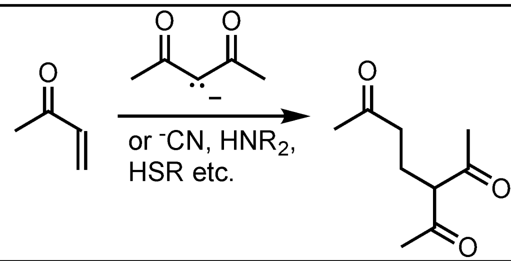
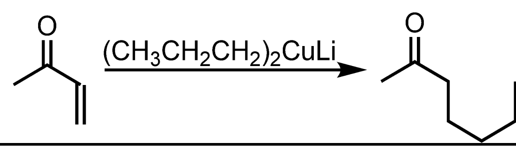
Acid-catalyzed Ring Opening of Epoxides (Nucleophile attacks more substituted side)	
Base-catalyzed Ring Opening of Epoxides (Nucleophile attacks less substituted side)	

Aldehyde and Ketone Reactions

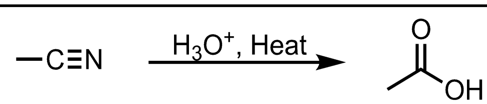
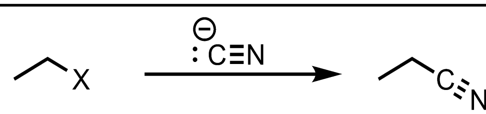
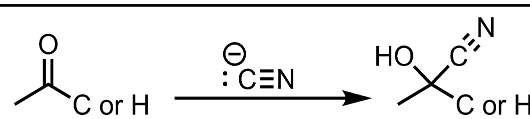
Nucleophilic Addition to an Aldehyde or Ketone	
Addition of water to an Aldehyde or Ketone forming a Hydrate	
Base-catalyzed addition of an Alcohol to an Aldehyde or Ketone forming a Hemi-acetal/Hemi-ketal	
Acid-catalyzed addition of an Alcohol to an Aldehyde or Ketone forming a Acetal/Ketal (Protecting Group, reversed by H_3O^+)	
Acid-catalyzed addition of Ethylene Glycol to an Aldehyde or Ketone forming a Acetal/Ketal (Protecting Group, reversed by H_3O^+)	
Addition of a 1˚ Amine to an Aldehyde or Ketone forming an Imine (Reversed by H_3O^+)	
Addition of a 2˚ Amine to an Aldehyde or Ketone forming an Enamine (Reversed by H_3O^+)	 Double bond forms on more substituted end for Ketones

$NaBH_4$

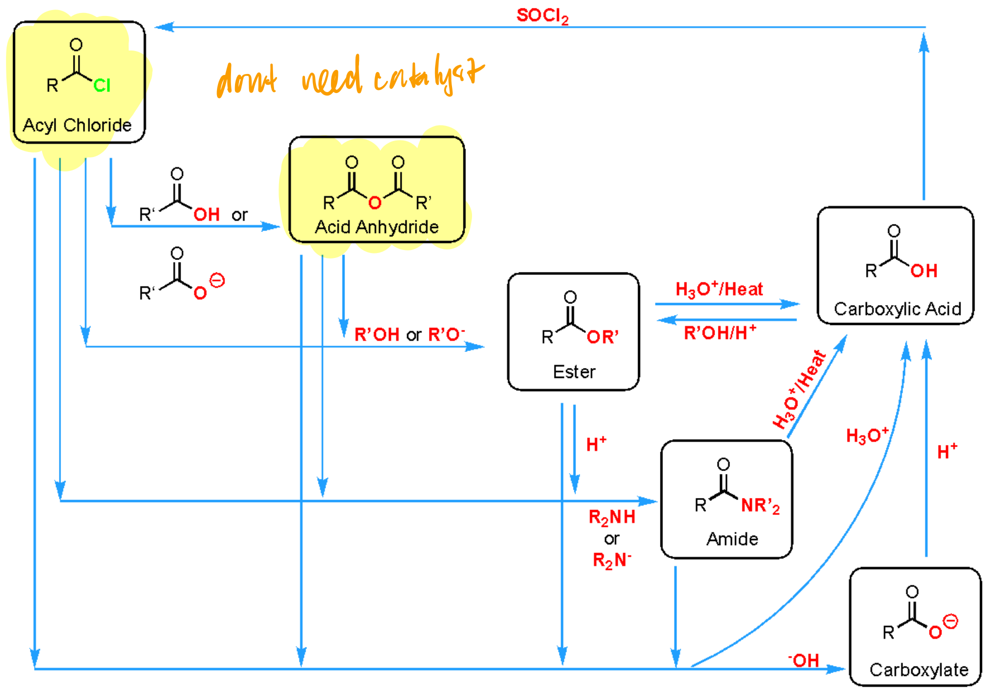
H_3O^+ only work on aldehydes & ketones

Addition of a Wittig Reagent to an Aldehyde/Ketone	
Michael Addition to an α, β Unsaturated Ketone	
Michael Addition to an α, β Unsaturated Ketone with a Gilman Reagent (Organocuprates)	

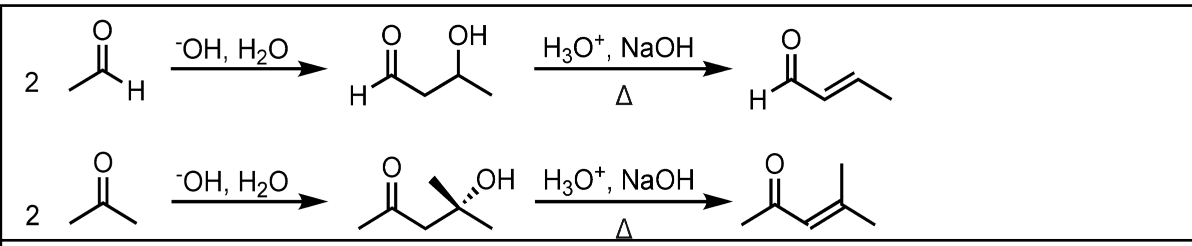
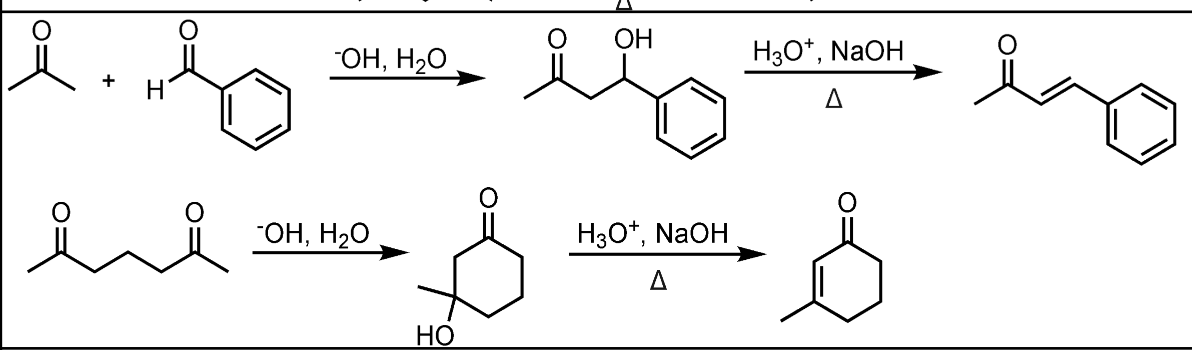
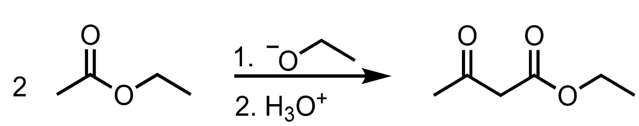
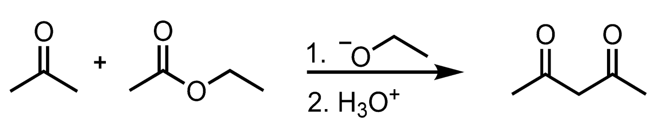
Nitrile Reactions

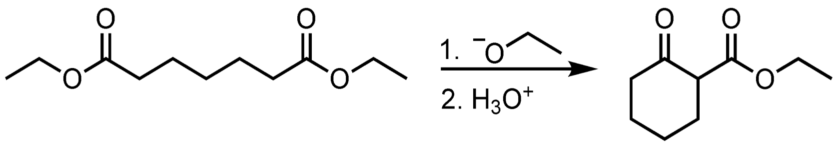
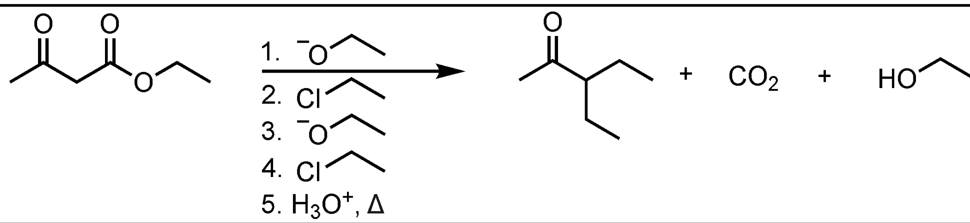
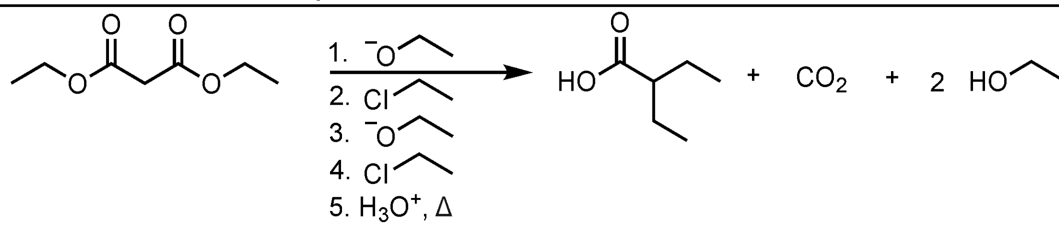
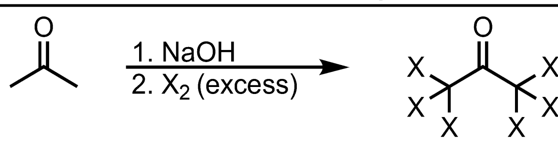
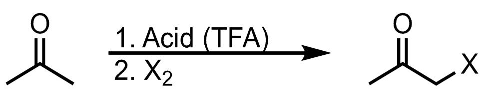
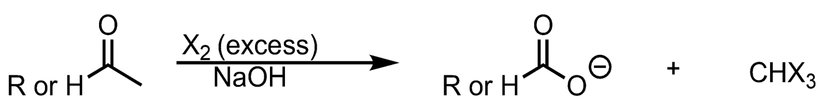
Acid-catalyzed Hydrolysis of a Nitrile	
SN2 formation of Nitriles using Cyanide and Alkyl Halides	
Cyanohydrin Formation using Aldehydes/Ketones and Cyanide	

Carboxylic Acid Derivative Reactions



Alpha Addition/Substitution Reactions

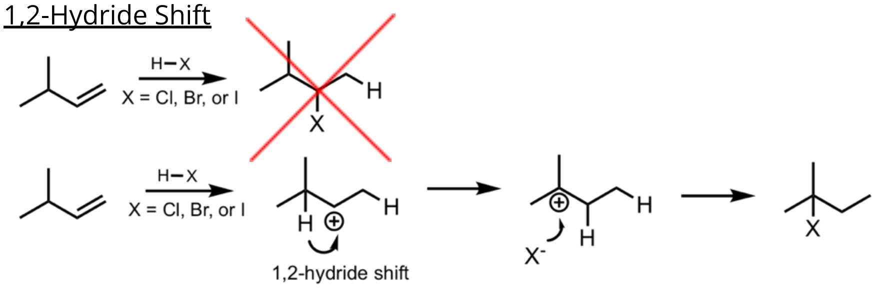
Self Aldol Condensation and Enone Formation	
Mixed Aldol Condensation and Enone Formation	
Self Claisen Condensation	
Mixed Claisen Condensation	

Dieckmann Cyclization (Intramolecular Claisen Condensation)	
Acetoacetic Ester Synthesis	
Malonic Ester Synthesis	
Alpha Halogenation In Basic Conditions	
Alpha Halogenation in Acidic Conditions	
Haloform Reaction	 ***A methyl group is required for this reaction**

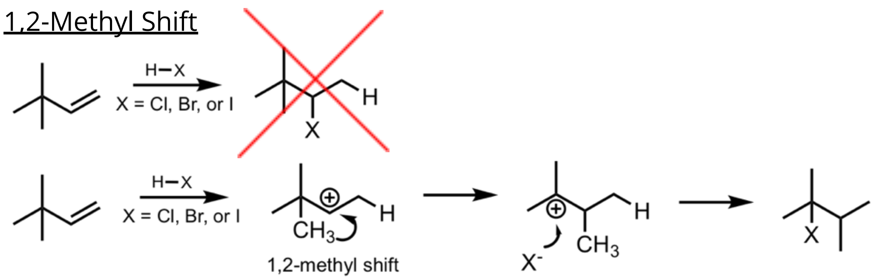
Rearrangements Details

When carbocations form, H's and CH_3's can do a 1,2-shift to generate a more stable carbocation intermediate

1,2-Hydride Shift

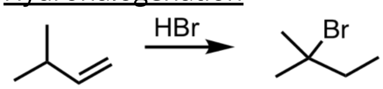


1,2-Methyl Shift

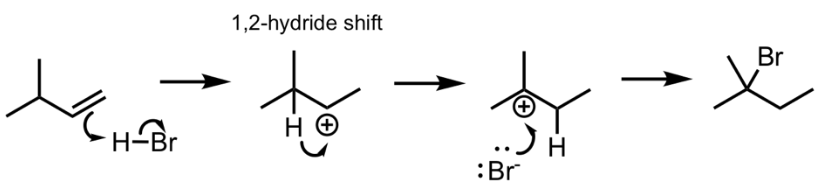


Alkene Reaction Details

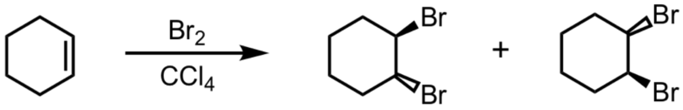
Hydrohalogenation



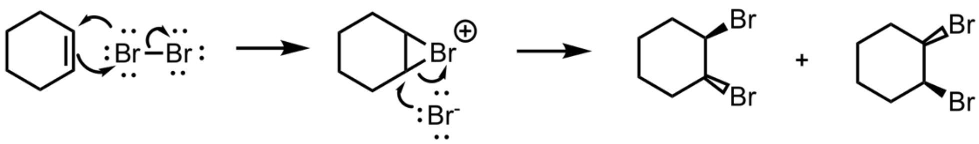
What's added: H+ and Br-
Regioselectivity: Markovnikov
Intermediate: Carbocation
Rearrangement: Possible (methyl and hydride shifts)
Mechanism:



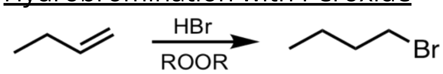
Halogenation



What's added: 2 Br atoms
Stereoselectivity: Anti
Intermediate: Bromonium ion
Rearrangement: Not possible
Mechanism:

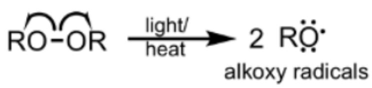


Hydrobromination with Peroxide

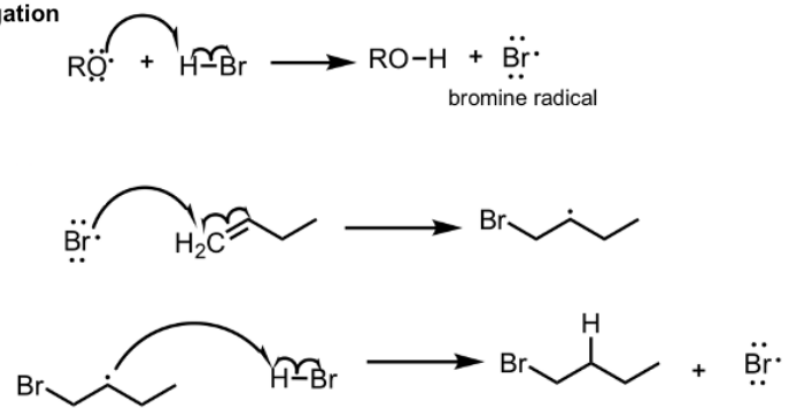


What's added: H· and Br·
Regioselectivity: Anti-Markovnikov
Intermediate: Radical
Rearrangement: Not possible
Mechanism:

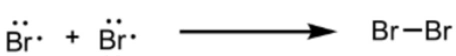
Step 1. Initiation



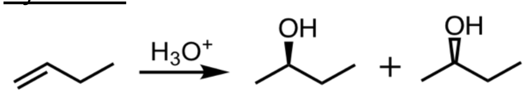
Step 2. Propagation



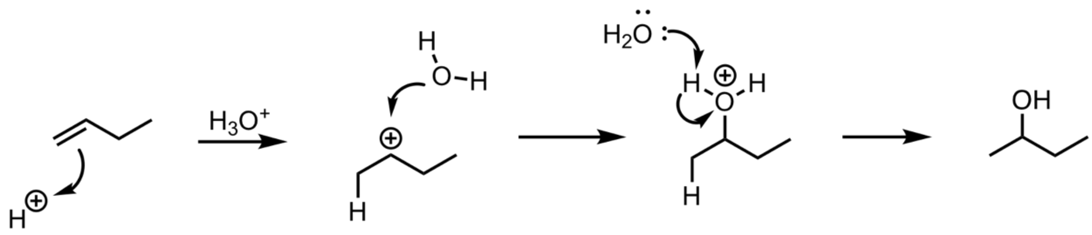
Step 3. Termination



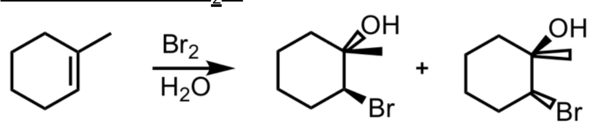
Hydration



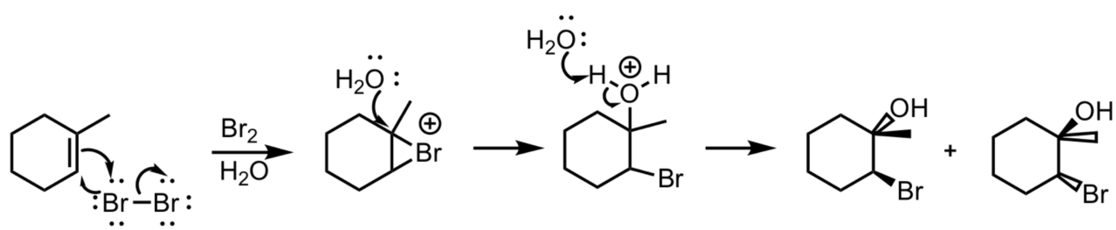
What's added: H^+ and OH^-
Regioselectivity: Markovnikov
Intermediate: Carbocation
Rearrangement: Possible (methyl and hydride shifts)
Mechanism:



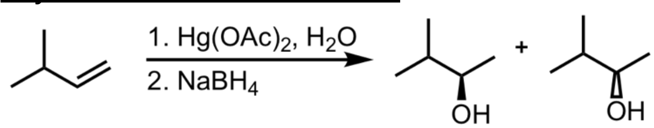
Bromination in H_2O



What's added: Br^+ and OH^-
Regioselectivity: Markovnikov
Stereoselectivity: Anti
Intermediate: Bromonium ion
Rearrangement: Not possible
Mechanism:

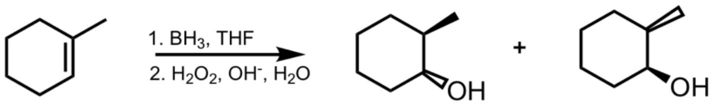


Oxymercuration-Demercuration



What's added: H^+ and OH^-
Regioselectivity: Markovnikov
Stereoselectivity: Anti
Intermediate: Mercurinium ion bridge
Rearrangement: Not possible

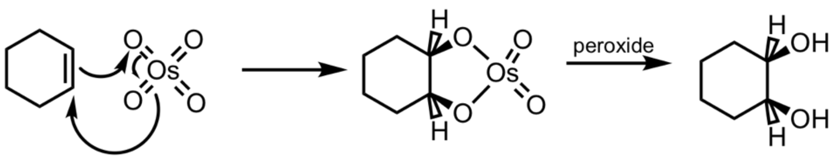
Hydroboration-Oxidation



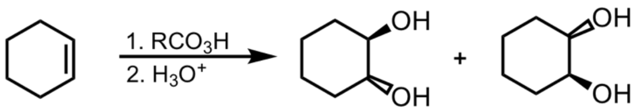
What's added: H^+ and OH^-
Regioselectivity: Anti-Markovnikov
Stereoselectivity: Syn
Intermediate: Hydroxy-boranes
Rearrangement: Not possible

Syn-Dihydroxylation

What's added: 2 OH groups
Stereoselectivity: Syn
Rearrangement: Not possible
Mechanism:

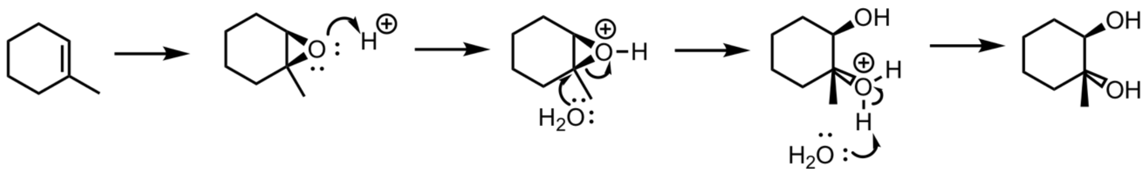


Anti-Dihydroxylation

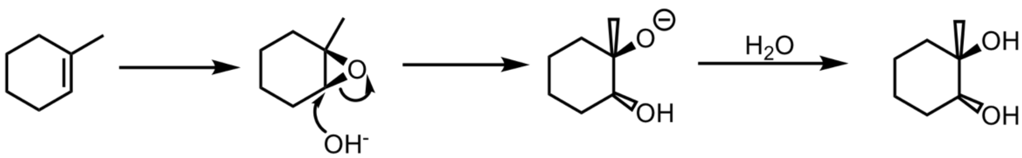


What's added: 2 OH groups
Stereoselectivity: Anti
Rearrangement: Not possible

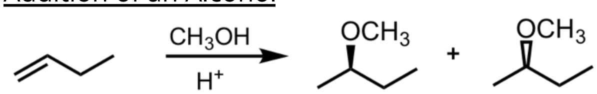
In acidic conditions, the H_2O attacks the more highly-substituted C:



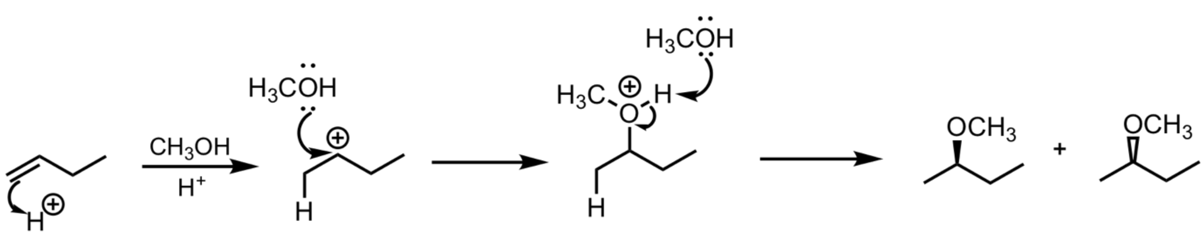
In basic conditions, H_2O attacks the less highly-substituted C:



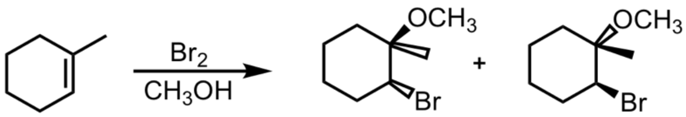
Addition of an Alcohol



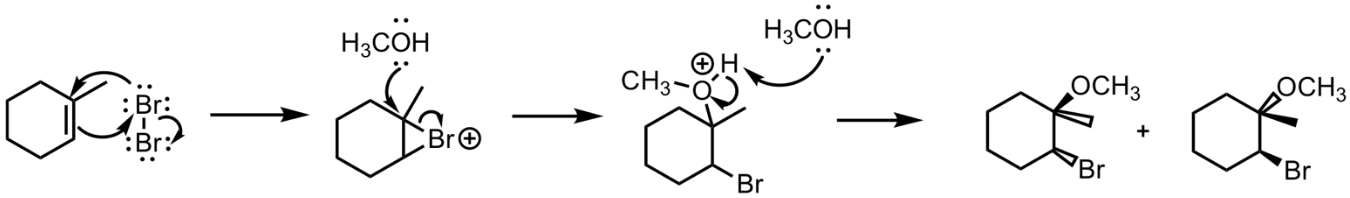
What's added: H^+ and OR^-
Regioselectivity: Markovnikov
Intermediate: Carbocation
Rearrangement: Possible
Mechanism:



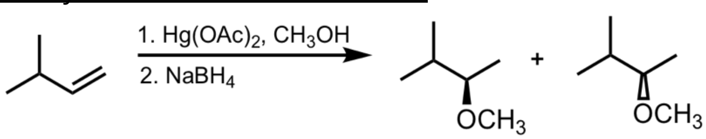
Bromination in Alcohol



What's added: Br^+ and OR^-
Regioselectivity: Markovnikov
Stereoselectivity: Anti
Intermediate: Bromonium ion
Rearrangement: Not possible
Mechanism:

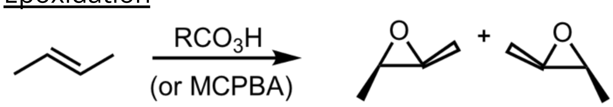


Alkoxymercuration-Demercuration

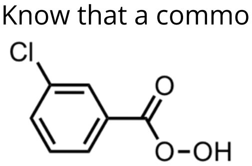


What's added: H^+ and OCH_3^-
Regioselectivity: Markovnikov
Stereoselectivity: Anti
Intermediate: Mercurinium ion
Rearrangement: Not possible

Epoxidation

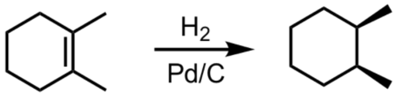


What's added: O
Stereoselectivity: Syn
Rearrangement: Not possible
Know that a commonly-used peroxy acid is *m*-CPBA:



m-CPBA

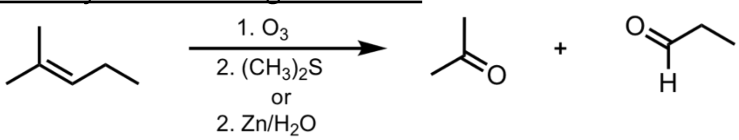
Catalytic Hydrogenation



meso

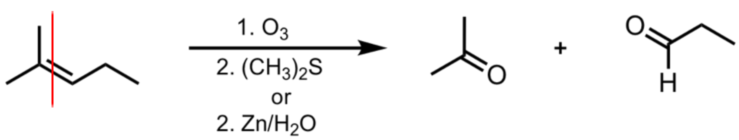
What's added: 2 H atoms
Stereoselectivity: Syn
Rearrangement: Not possible
Note: You may see Pt used as well. This is just the catalyst and does not change the outcome of the products.

Ozonolysis in Reducing Conditions

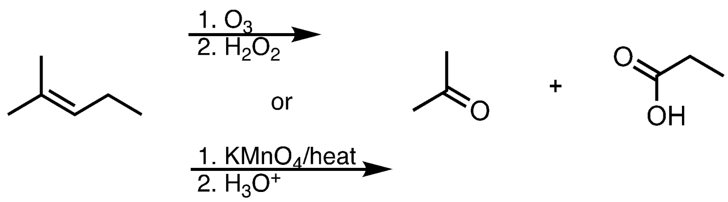


What's added: 2 O atoms
Regioselectivity: N/A
Stereoselectivity: N/A
Intermediate: N/A
Rearrangement: N/A
Mechanism: *You do not need to know the mechanism for this reaction*

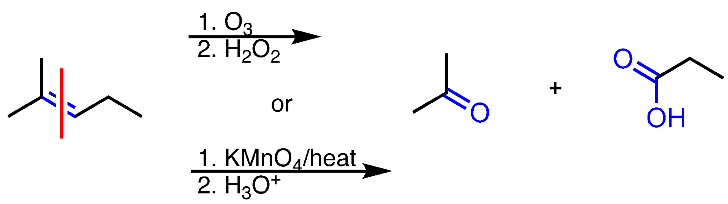
Know that the C=C double bond gets "sawed" in half, and an O atom is placed on the end of each new piece.
Note: $(CH_3)_2S$ is often abbreviated "DMS" for dimethyl sulfide.



Ozonolysis in Oxidizing Conditions/Oxidative Cleavage



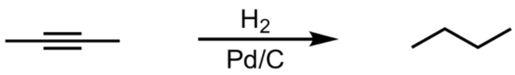
What's added: Multiple O atoms
Regioselectivity: N/A
Stereoselectivity: N/A
Intermediate: N/A
Rearrangement: N/A
Mechanism: *You do not need to know the mechanism for this reaction*



Know that the C=C double bond gets "sawed" in half, and an O atom is placed on the end of each new piece. Any H's attached to the alkene C's get replaced by an –OH group since we are under oxidizing conditions/hot $KMnO_4$. Unlike reducing conditions which would have formed aldehydes, oxidizing conditions produce carboxylic acids instead.

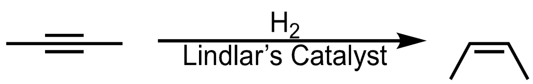
Alkyne Reaction Details

Catalytic Hydrogenation



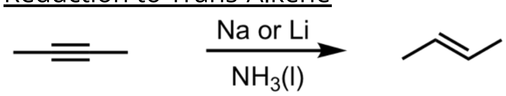
What's added: 4 H atoms
Stereoselectivity: Anti
Rearrangement: Not possible
Note: You may see Pt used as well. This is just the catalyst and does not change the outcome of the products.

Reduction to Cis-Alkene



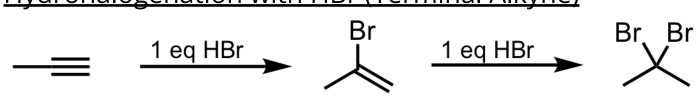
What's added: 2 H atoms
Stereoselectivity: Syn
Rearrangement: Not possible

Reduction to Trans-Alkene

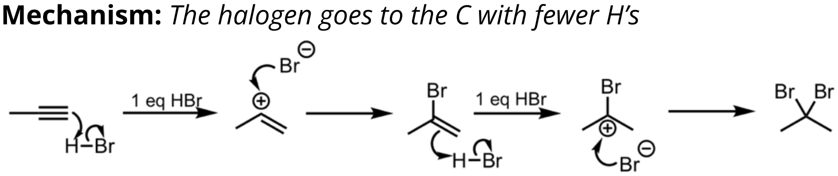


What's added: 2 H atoms
Stereoselectivity: Anti
Rearrangement: Not possible

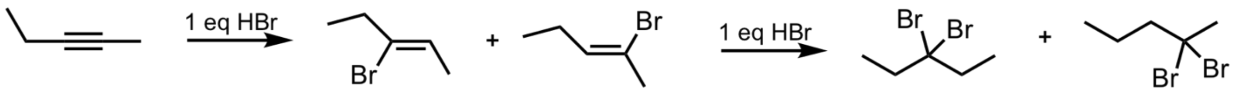
Hydrohalogenation with HBr (Terminal Alkyne)



What's added: 1 H atom and 1 halogen atom (can be F, Br, I, or Cl) per equivalent of HX
Regioselectivity: Markovnikov
Intermediate: Carbocation
Rearrangement: Not possible
Mechanism: *The halogen goes to the C with fewer H's*

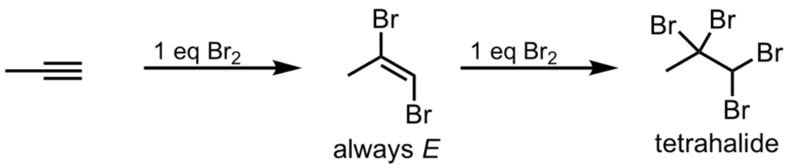


Hydrohalogenation with HBr (Internal Alkyne)

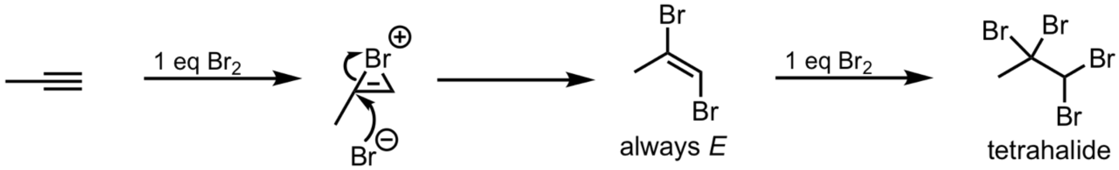


What's added: 1 H atom and 1 halogen atom (can be Cl or Br) per equivalent of HX
Regioselectivity: Markovnikov
Intermediate: Carbocation
Rearrangement: Not possible
Mechanism: *Same as for terminal alkynes, but yields a mixture of two products because both intermediates are equally stable*

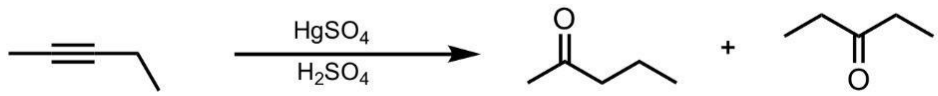
Halogenation with Br_2



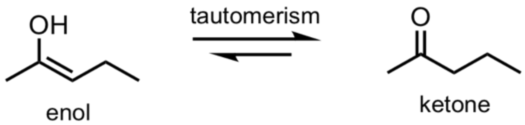
What's added: 2 halogen atoms (can be F, Br, I, or Cl) per equivalent of X_2
Stereoselectivity: Anti
Intermediate: Bromonium ion
Rearrangement: Not possible
Mechanism:



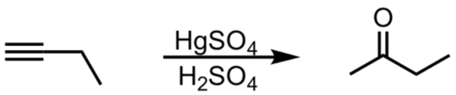
Hydration of an Internal Alkyne



What's added: 1 O atom
Rearrangement: Not possible
Do know that this reaction produces enols, which then tautomerize to form ketones.

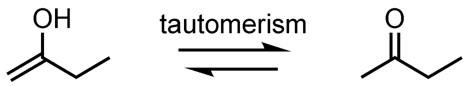


Hydration of a Terminal Alkyne (Markovnikov)



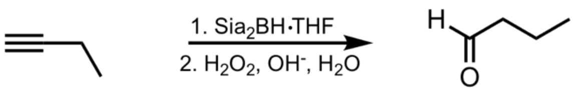
What's added: 1 O atom
Regioselectivity: Markovnikov
Rearrangement: Not possible

Know that this reaction produces Markovnikov enols, which then tautomerize to form ketones.



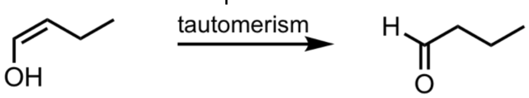
enol (Markovnikov) ketone

Hydration of a Terminal Alkyne (Anti-Markovnikov)



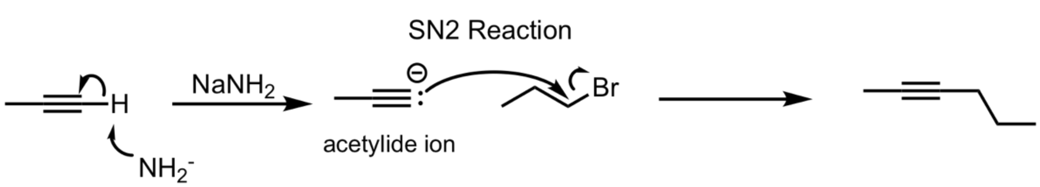
What's added: 1 O atom
Regioselectivity: Anti-Markovnikov
Rearrangement: Not possible

Know that this reaction produces Anti-Markovnikov enols, which then tautomerize to form aldehydes.



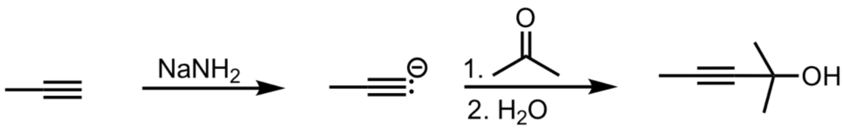
enol (Anti-Markovnikov) aldehyde

SN2 Addition of an Acetylide Ion to an Alkyl Halide



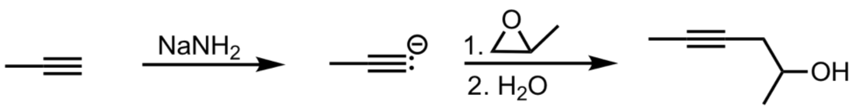
What's added: additional C atoms (-R of alkyl halide)
Intermediate: Acetylide Ion
Rearrangement: Not possible
Mechanism: Deprotonation, then alkylation via SN2 reaction

Addition of an Acetylide Ion to a Ketone

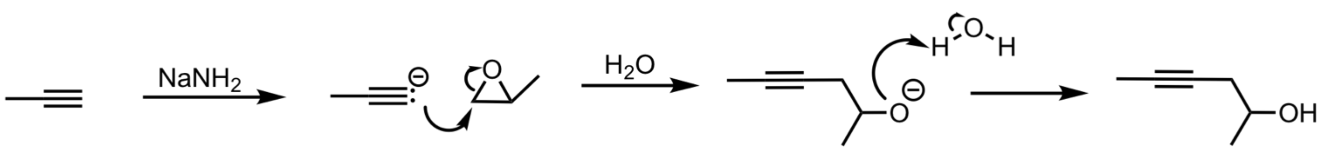


What's added: 1 alkyl group
Intermediate: Acetylide Ion
Rearrangement: Not possible
Mechanism: Deprotonation, then addition of a ketone

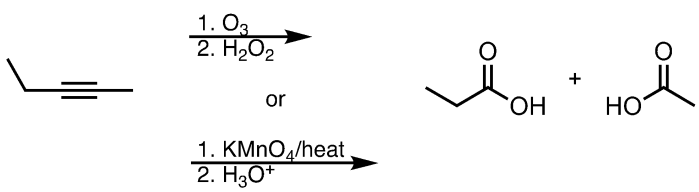
SN2 Addition of an Acetylide Ion to an Epoxide



What's added: 2-hydroxylpropane (from epoxide)
Intermediate: Acetylide Ion
Rearrangement: Not possible
Mechanism: Deprotonation, then addition of 2-hydroxyl propane via SN2 reaction

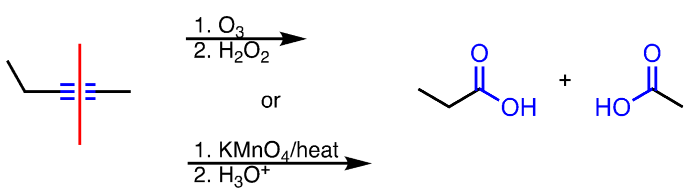


Ozonolysis/Oxidative Cleavage on an Internal Alkyne

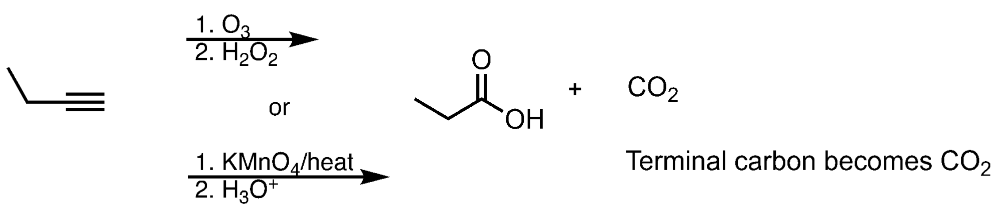


What's added: 4 O atoms and 2 H atoms

Know that the reaction cuts the triple bond in half. An O replaces two of the bonds as C=O and the third lone bond becomes a bond to -OH.

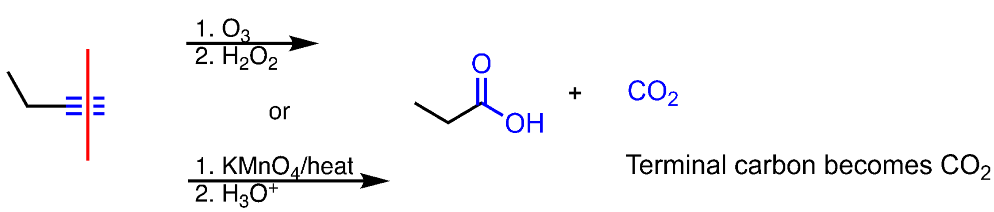


Ozonolysis/Oxidative Cleavage on a Terminal Alkyne



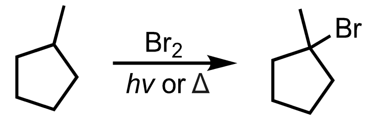
What's added: 4 O atoms and 1 H atom

Know that the reaction cuts the triple bond in half. On the internal side, an O replaces two of the bonds as C=O and the third lone bond becomes a bond to -OH. On the terminal side, two oxygens O replace all the bonds on carbon, forming the most oxidized form of carbon: CO_2.



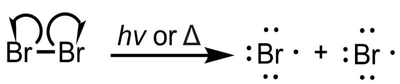
Free Radical Halogenation Reaction Details

<u>Free Radical Halogenation using Bromine (more selective)</u>

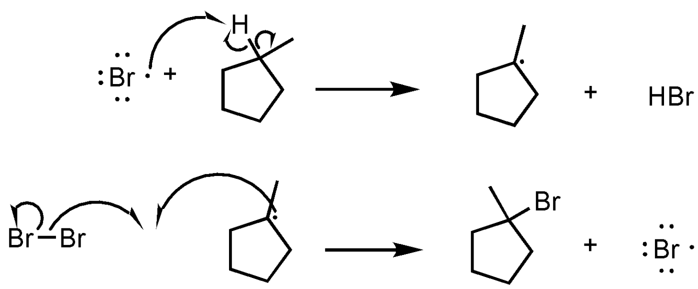


What's added: 1 Br atom
Regioselectivity: Most Substituted Product
Intermediate: Radical Intermediate
Rearrangement: Not possible
Mechanism: Formation of bromine and carbon radicals and them joining to create an alkyl halide

1. Initiation

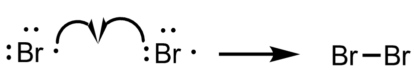


2. Propagation

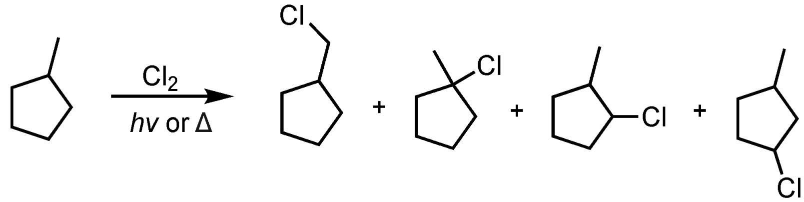


alkyl halide

3. Termination

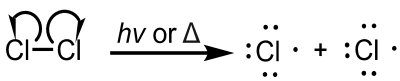


Free Radical Halogenation using Chlorine (less selective)

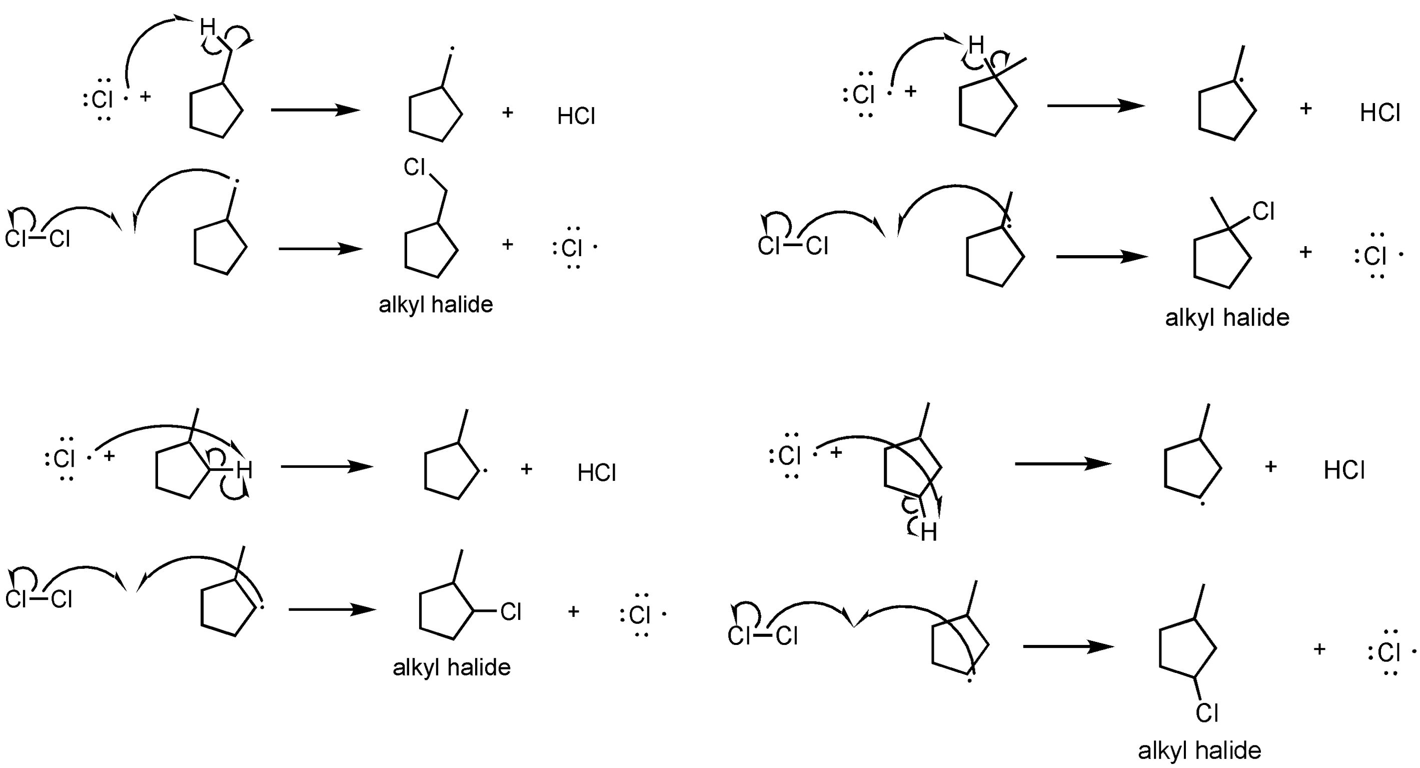


What's added: 1 Cl atom
Intermediate: Radical Intermediate
Rearrangement: Not possible
Mechanism: Formation of chlorine and carbon radicals and them joining to create alkyl halides

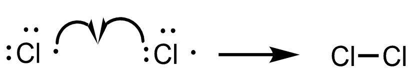
1. Initiation



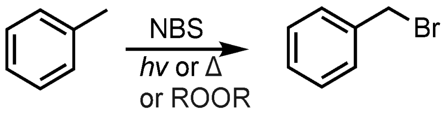
2. Propagation

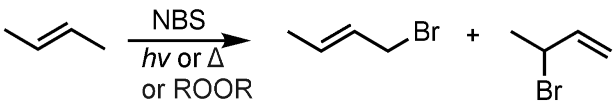


3. Termination



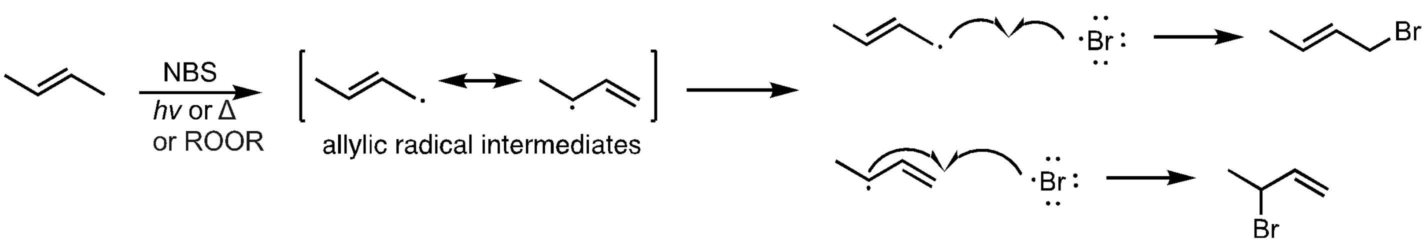
Allylic/Benzylic Bromination





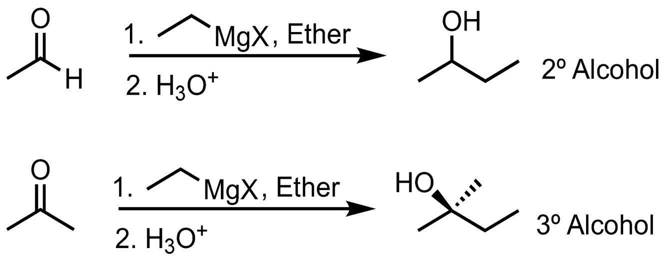
What's added: 1 Br atom
Intermediate: Allylic Radical Intermediate
Rearrangement: Not possible

Note: this reaction results in the formation of allylic radical intermediates which resonate and thus allow for the formation of multiple products.

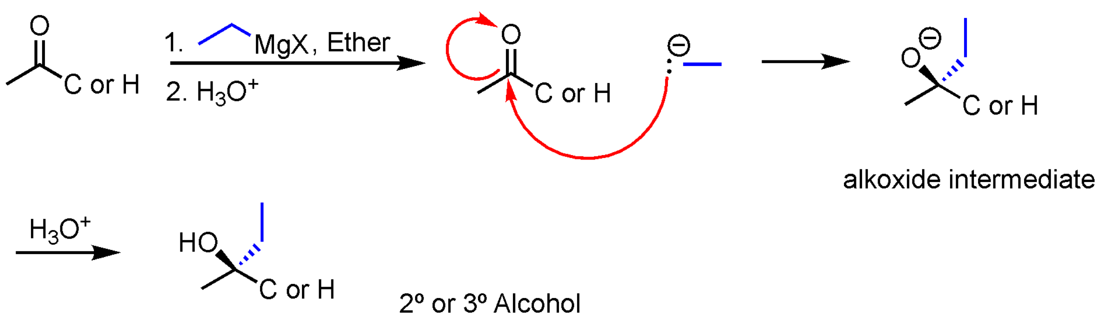


Grignard Reaction Details

Addition of a Grignard Reagent to an Aldehyde/Ketone

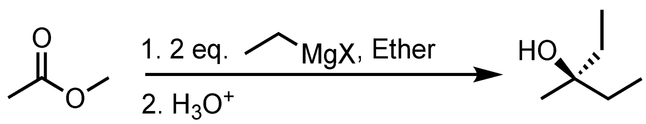


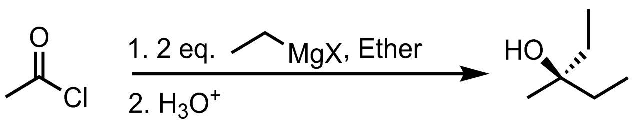
What's added: R group
Intermediate: Alkoxide ion
Mechanism: Grignard reagents are electron-rich and attack electrophilic species such as carbonyls. The mechanism follows typical nucleophilic addition to a carbonyl, forming 2° or 3° Alcohols.



Note: Using R-MgX is virtually the same reaction as using R-Li (organolithium); you are using the R-group as a nucleophile to form an alcohol.

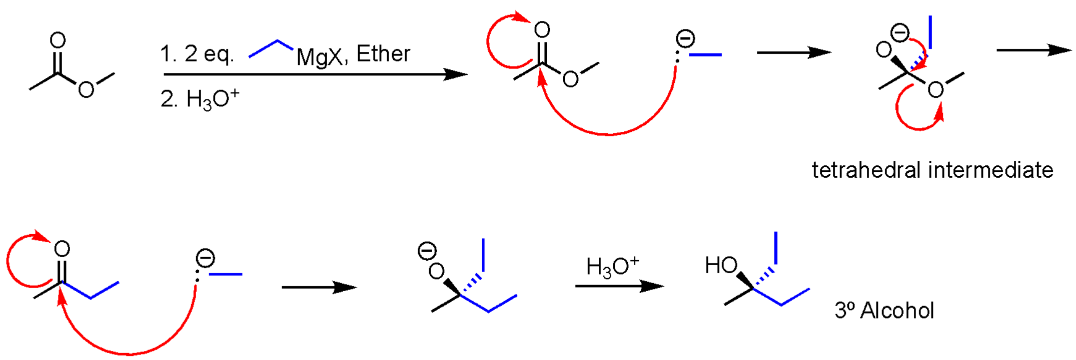
Addition of a Grignard Reagent to an Ester/Acyl Chloride





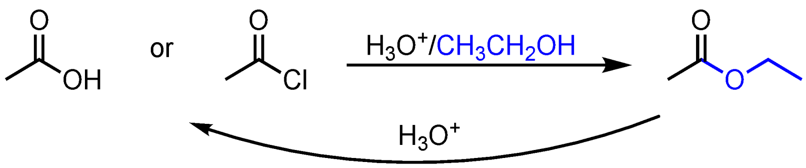
What's added: 2 R groups

Mechanism: Grignard reagents are electron-rich and attack electrophilic species such as carbonyls. The mechanism follows typical nucleophilic addition to a carbonyl with a twist; the Grignard reagent adds **twice** with esters and acyl chlorides, forming 3º alcohols. Here it is shown on an ester (the same mechanism as with an acyl chloride):

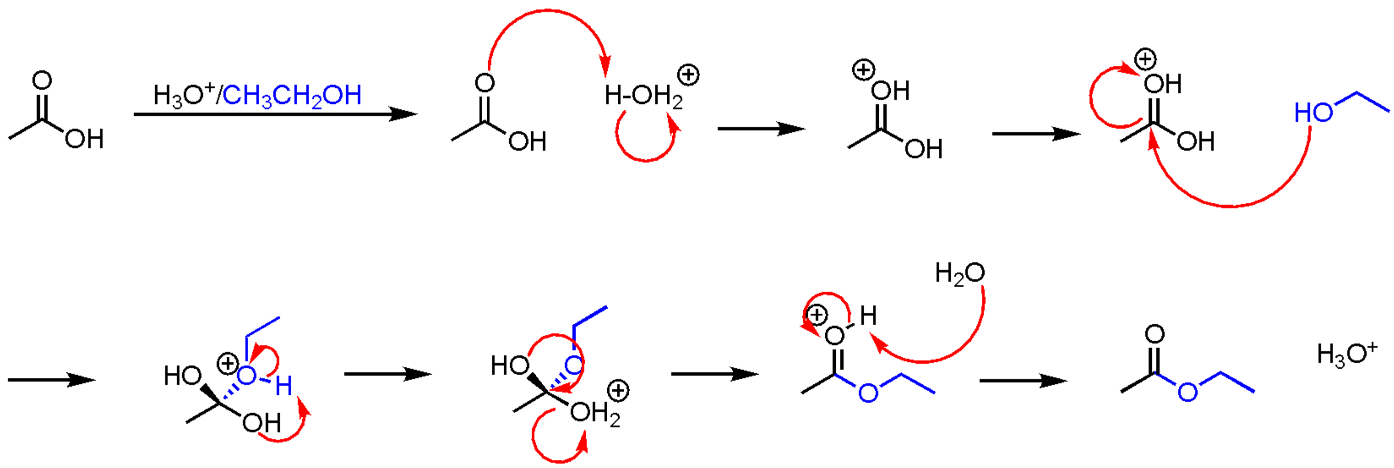


Carboxylic Acid Derivative Reaction Details

<u>Fischer Esterification</u>



What's added: An alcohol
Mechanism: Essentially, this reaction substitutes the Cl or OH of an acyl chloride/ester with an alcohol:



tetrahedral intermediate

Notes: Notice that the reaction is reversible under H_3O^+/heat. The reverse reaction is how esters are cleaved into their corresponding carboxylic acid and alcohol, an important biological process. Also, this is the same mechanism for any reaction involving a carboxylic acid derivative (anhydrides, amides, carboxylic acids, esters).